

2020 AMC 10A Problem 7 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10A Problem 7. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10A Problem 7

Problem:

The 25 integers from -10 to 14 , inclusive, can be arranged to form a 5 -by- 5 square in which the sum of the numbers in each row, the sum of the numbers in each column, and the sum of the numbers along each of the main diagonals are all the same. What is the value of this common sum?

Answer Choices:

- A. 2
- B. 5
- C. 10
- D. 25
- E. 50

Solution:

Without loss of generality, consider the five rows in the square. Each row must have the same sum of numbers, meaning that the sum of all the numbers in the square divided by 5 is the total value per row. The sum of the 25 integers is

$$-10 + -9 + \dots + 14 = 11 + 12 + 13 + 14 = 50$$

and the common sum is

$$\frac{50}{5} = \boxed{(C) 10}$$

OR

Take the sum of the middle 5 values of the set (they will turn out to be the mean of each row). We get

$$0 + 1 + 2 + 3 + 4 = \boxed{(C) 10}.$$

as our answer.

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